Early Prediction of Heart Disease using ML

Data Science Course Project

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***Abstract*— Heart disease is one of the main reasons for death worldwide, making early prediction critical for reducing deaths. Current diagnostic methods are often expensive and slow. This study applies data science techniques to predict heart disease using the UCI Cleveland Heart Disease dataset obtained from the UCI Machine Learning Repository, which contains 303 patient records with 14 clinical features. The objectives of this study are to preprocess the dataset, identify significant predictors of heart disease, and develop accurate classification models. Data preprocessing steps were performed including handling missing values in the ca (number of major vessels) and thal (thalassemia type) columns using median imputation. Exploratory data analysis confirmed that maximum heart rate (thalach) is the strongest predictor of heart disease, showed by a clear t-test result ($p<0.05$).**

Keywords—heart disease, machine learning, EDA, UCI Cleveland

I. Introduction

Heart disease is one of the most serious health conditions in the world, with around 1 million deaths in the United States in 2022 [1]. Cardiovascular disease is responsible for nearly 32% of all global deaths yearly [2], which makes it a critical public health concern. Early detection of heart disease is very important to lower death rates and improve patient outcomes. However, the complexity of cardiovascular conditions makes accurate diagnosis challenging, requiring multiple clinical tests and specialist consultations.

Traditional diagnostic methods for heart disease are often slow, expensive, and may result in late detection which can be life threatening. This study addresses this problem by applying data science techniques to the UCI Cleveland Heart Disease dataset to automatically predict heart disease. The primary objectives are to preprocess the dataset, identify the most significant clinical predictors of heart disease, and evaluate machine learning classification models for accurate prediction. Applying data science techniques to this problem is significant as it provides a faster and more cost-effective alternative to traditional methods, ultimately assisting clinicians in making earlier and more accurate decisions, reducing mortality rates and improving patient outcomes.

II. LITERATURE REVIEW

Several studies have explored the application of machine learning techniques for heart disease prediction, showing good outcomes in enhancing diagnostic accuracy. Various ML approaches have been tested, including Logistic Regression, Random Forest, Support Vector Machine, and Neural Networks, each proving potential in supporting clinicians with early detection of cardiovascular disease. The following review covers key studies in this area, focusing on their methodologies, findings, and limitations.

Bhatt et al. [3] proposed a machine learning model to predict cardiovascular disease using a dataset of 70k patient records collected from Kaggle. The study applied four classification algorithms including Random Forest, Decision Tree, XGBoost, and Multilayer Perceptron, with hyperparameter tuning to optimize results. MLP achieved the highest accuracy of 87.28%, surpassing other models. However, the study did not handle class imbalance, which may impact prediction performance on minority classes.

A study published on IEEE Xplore [4] evaluated four classification methods including MLP, SVM, Random Forest, and Naïve Bayes for heart disease prediction using the UCI Cleveland dataset. Data preprocessing and feature selection were performed before model training. SVM achieved the best performance with 91.67% accuracy. However, the study was limited by the small size of the Cleveland dataset which has only 303 records.

Another IEEE study [5] focused on early stage heart disease prediction using feature selection methods including chi-square, ANOVA, and mutual information to identify the most relevant clinical features. The study showed that proper feature selection significantly improves model accuracy. However, the study was limited to a single dataset, reducing the applicability of the findings.

A study published in Scientific Reports [6] developed a heart disease prediction model combining genetic algorithm and recursive feature elimination for feature selection, along with a deep convolutional neural network for classification. The model handled class imbalance using an under-sampling clustering oversampling method. The developed model showed outstanding performance in multiple evaluation metrics. However, the high computational complexity of the model limits its practical deployment in clinical settings.

A comprehensive review published in Frontiers in Artificial Intelligence [7] examined machine learning approaches for heart disease prediction, identifying key challenges and future directions. The review found that ensemble methods generally outperform single classifiers in terms of accuracy and robustness.

The study also highlighted class imbalance as a major challenge in heart disease datasets. However, as a review paper, it did not propose or evaluate a new prediction model.

TABLE I. Comparison of Heart Disease Prediction Studies

Study	ML Models	Dataset	Key Results	Limitations
Bhatt et al. [3]	RF, DT, XGBoost, MLP	Kaggle (70k)	MLP: 87.28%	Class imbalance not addressed
IEEE [4]	MLP, SVM, RF, NB	UCI Cleveland	SVM: 91.67%	Small dataset

IEEE [5]	Chi-square, ANOVA, MI	UCI Cleveland	Improved accuracy	Single dataset only
Sci. Reports [6]	GA, CNN	UCI Cleveland	Outstanding performance	High computational cost
Frontiers [7]	Various	Multiple	Ensemble methods best	No new model proposed

III. Dataset and Methodology

A. Data Collection, Description, and Preprocessing

The dataset utilized in this study was obtained from the UCI Machine Learning Repository, specifically the Cleveland Heart Disease dataset [8]. This dataset was selected due to its extensive use as a benchmark in cardiovascular research, enabling meaningful comparison with prior studies [3,4,5,6]. The dataset comprises 303 patient records with 14 clinical features including age, sex, chest pain type (cp), resting blood pressure (trestbps), serum cholesterol (chol), fasting blood sugar (fbs), resting ECG results (restecg), maximum heart rate (thalach), exercise induced angina (exang), ST depression (oldpeak), slope, number of major vessels (ca), thalassemia (thal), and the target variable indicating heart disease presence.

A known limitation of this dataset is its relatively small size of 303 records, which may impact the applicability of the trained models. Additionally, the dataset was collected from a single medical institution, which may introduce selection bias and limit applicability to broader populations.

B. Data Science Techniques Overview

Data preprocessing involved several key steps. First, missing values were identified in the ca column (4 missing) and thal column (2 missing), representing less than 2% of the total records. These were filled using median imputation to preserve data integrity and avoid the influence of outliers. Second, the target variable was converted from multi-class values (0-4) to binary format, where 0 indicates no heart disease and 1 indicates presence of heart disease, making it suitable for binary classification. Third, exploratory data analysis was performed including descriptive statistics, outlier detection using z-score method, and correlation analysis to identify significant predictors.

The analysis was implemented using Python programming language with the following libraries: Pandas for data manipulation, NumPy for numerical computations, Matplotlib and Seaborn for data visualization, and SciPy for statistical hypothesis testing. All code was developed and executed in DataSpell IDE.

IV. Preliminary Analysis

A. Initial Steps

After loading the UCI Cleveland Heart Disease dataset, an initial inspection was performed to assess data quality. The dataset contains 303 patient records and 14 clinical features, with the target variable converted from multi-class values (0-4) to binary format where 0 indicates no heart disease and 1 indicates presence of heart disease. Missing values were detected in two columns: ca (number of major vessels) with 4 missing values and thal (thalassemia type) with 2 missing values, representing less than 2% of the total records. These missing values were handled using median imputation to maintain data integrity and avoid the influence of outliers on the imputed values.

Initial inspection also showed that the dataset has a near-balanced target distribution with 54.1% no disease and 45.9% disease, and that most patients are male (68%) with a mean age of 54.44 years, indicating that middle-aged males represent the highest risk group in this dataset.

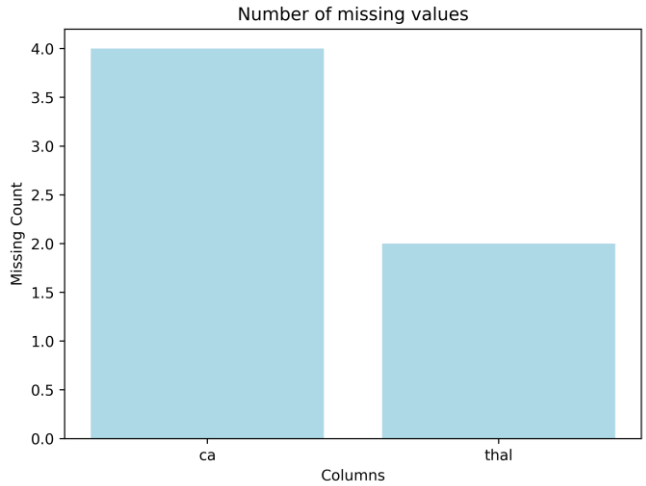


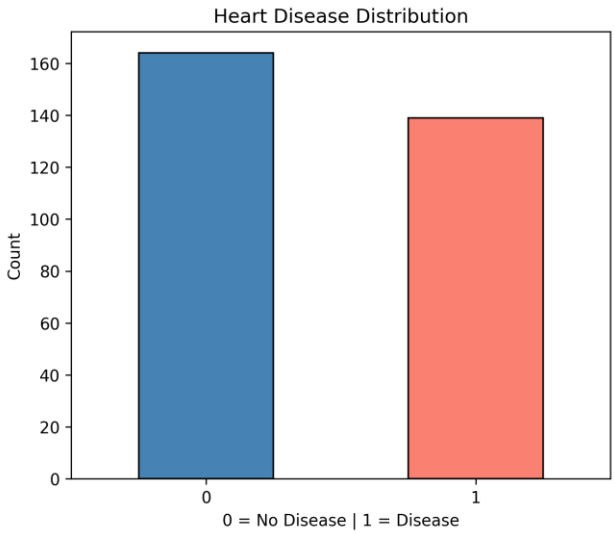
Fig. 1. Number of Missing Values per Column

B. Data Exploration

Descriptive statistical analysis was performed to describe the averages and distributions of all numerical features. The mean age of patients is 54.44 years (SD=9.04), ranging from 29 to 77 years. The mean cholesterol level is 246.69 mg/dl (SD=51.78), with a maximum value of 564 mg/dl indicating potential outliers. The mean maximum heart rate (thalach) is 149.61 (SD=22.88). The target variable shows a near-balanced distribution with 164 patients (54.1%) showing no heart disease and 139 patients (45.9%) diagnosed with heart disease, making it suitable for machine learning classification.

Fig. 2. Heart Disease Distribution

Outlier analysis was performed using the z-score method with a threshold of $|z| > 3$. Results showed that the cholesterol column had the highest number of outliers (4 outliers), including one extreme value of 564 mg/dl. Additionally,



trestbps and oldpeak each contained 2 outliers, while thalach had 1 outlier. Since these values are medically plausible, they were retained in the dataset to preserve data integrity.

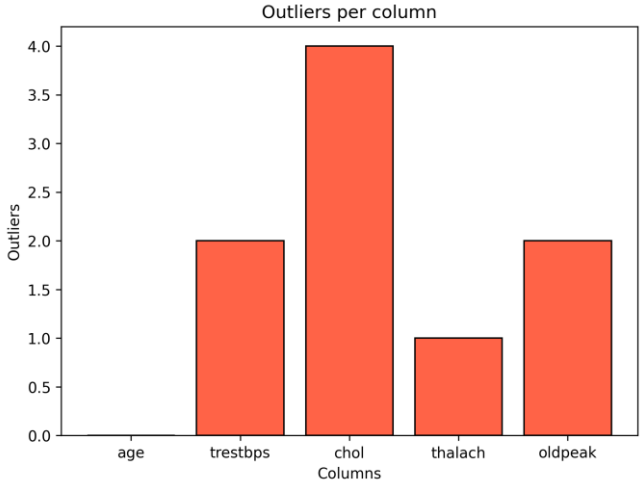


Fig. 3. Number of Outliers per Column

Histogram analysis was performed to visualize the distribution of key numerical features. The age distribution shows that most patients are between 50-65 years old, with a relatively normal distribution. The cholesterol distribution is right-skewed, showing the presence of high outlier values. The maximum heart rate distribution looks mostly normal, while resting blood pressure shows a slight right skew.

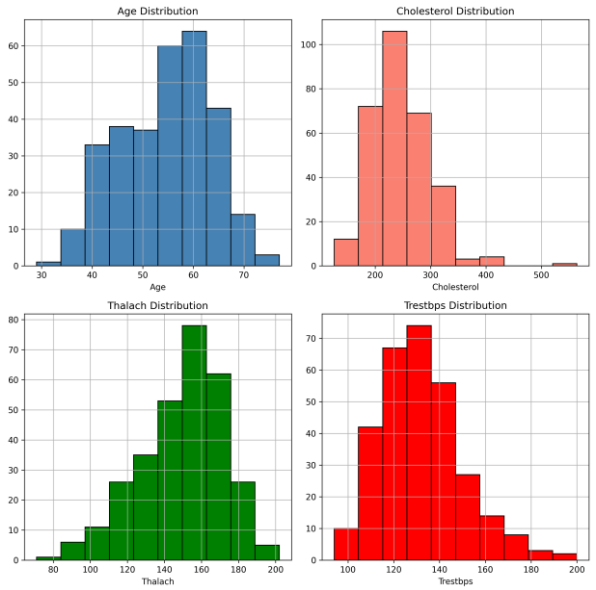


Fig. 4. Distribution of Key Numerical Features

Correlation analysis was performed to identify relationships between clinical features and heart disease. As shown in Figure 5, thal (0.52) and ca (0.46) showed the strongest positive correlation with heart disease, while thalach (-0.42) showed the strongest negative correlation, indicating that patients with higher maximum heart rates are less likely to have heart disease. Interestingly, cholesterol showed minimal correlation (0.09) with the target variable.

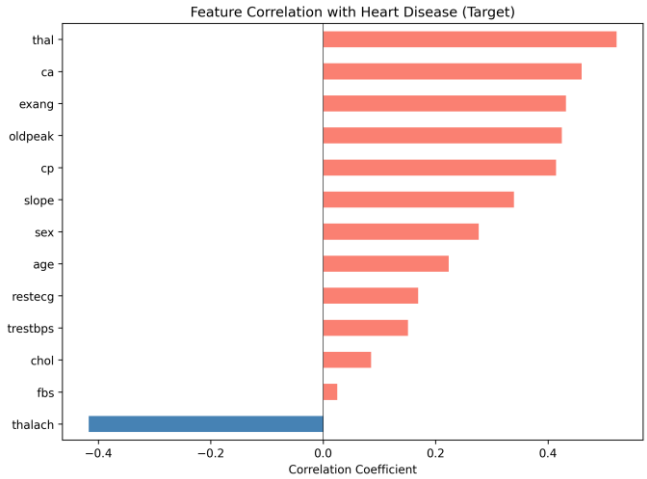


Fig. 5. Feature Correlation with Heart Disease

C. *Visualizations and Findings*

Two hypotheses were developed and tested using independent samples t-tests to examine significant differences between patients with and without heart disease.

Hypothesis 1: There is a significant difference in cholesterol levels between patients with and without heart disease.

- No Disease mean: 242.64 mg/dl
- Has Disease mean: 251.47 mg/dl
- T-statistic: -1.48, p-value: 0.1391
- Result: $p > 0.05$ → **Not significant** cholesterol alone is not a strong predictor

Hypothesis 2: There is a significant difference in maximum heart rate between patients with and without heart disease.

- No Disease mean: 158.38
- Has Disease mean: 139.26
- T-statistic: 7.96, p-value: 0.0000
- Result: $p < 0.05$ → **Significant** maximum heart rate is a strong predictor

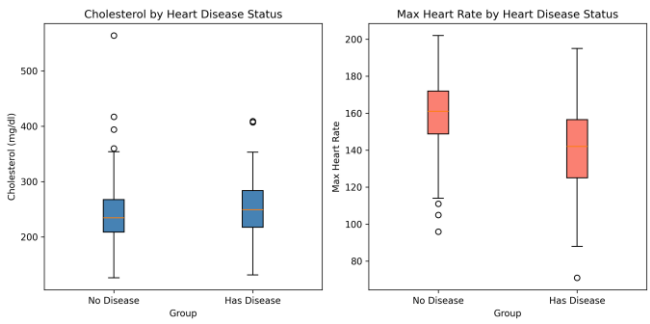


Fig. 6. Cholesterol and Max Heart Rate by Heart Disease Status

The boxplot in Figure 6 visually confirms the t-test findings. The cholesterol boxplots show significant overlap between groups, consistent with the non-significant t-test result. In contrast, the maximum heart rate boxplots show clear separation between groups, confirming its role as a strong predictor of heart disease.

V. Conclusion

This study presented a preliminary analysis of the UCI Cleveland Heart Disease dataset with the objective of developing an accurate prediction model for cardiovascular

disease. The dataset containing 303 patient records with 14 clinical features was preprocessed by handling missing values in the ca and thal columns using median imputation. Exploratory data analysis showed that thal, ca, and exang are the most significant positive predictors of heart disease, while maximum heart rate (thalach) showed the strongest negative correlation (-0.42). Hypothesis testing confirmed a clear difference in maximum heart rate between diseased and healthy patients ($p<0.05$), while cholesterol showed no significant difference ($p=0.1391$).

In Milestone 2, three machine learning classification models including Logistic Regression, Decision Tree, and Random Forest will be trained and evaluated on the preprocessed dataset. Model performance will be measured using accuracy, precision, recall, and F1-score metrics to identify the most accurate approach for heart disease prediction, with the goal of providing a robust and efficient clinical decision support tool.

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